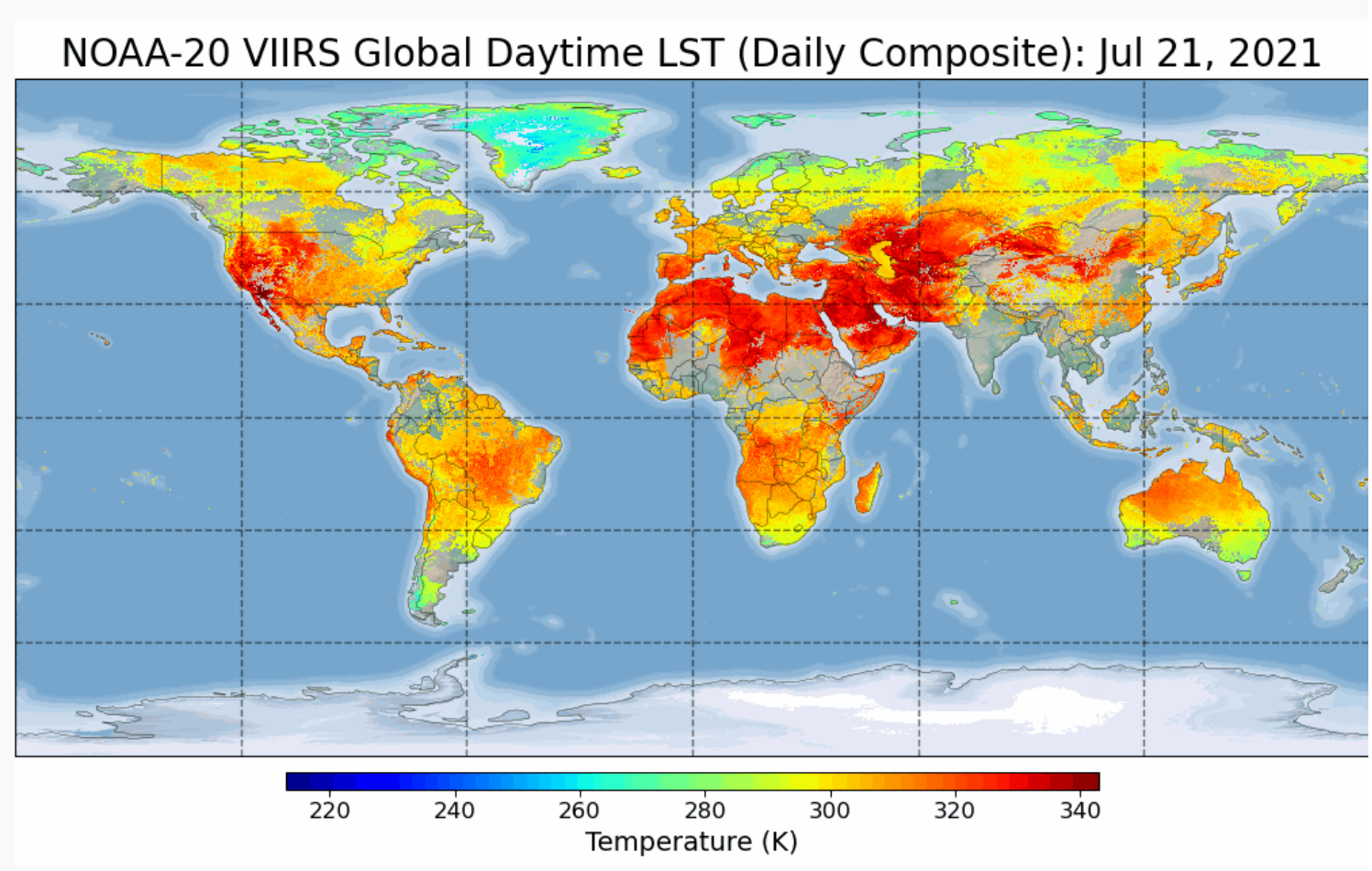
Passive remote sensing: elements of thermodynamics



VIIRS (Visible Infrared Imaging Radiometer Suite) aboard the joint NASA/NOAA Suomi National Polar-orbiting Partnership

Remote sensing: basic laws

EM Remote sensing measures an EM field and tries to infer the physical properties of the object that generated or reflected it, at the finest possible resolution. The properties of the emitted or reflected field are of **utmost** importance at each wavelength

The goal is to estimate the (energy) of the emitted / reflected field of the 'target', for each λ.

We also need to know if the EM signature is stronger than the noise (sensitivity).

We have four fundamental laws, ruling the energy spectrum:

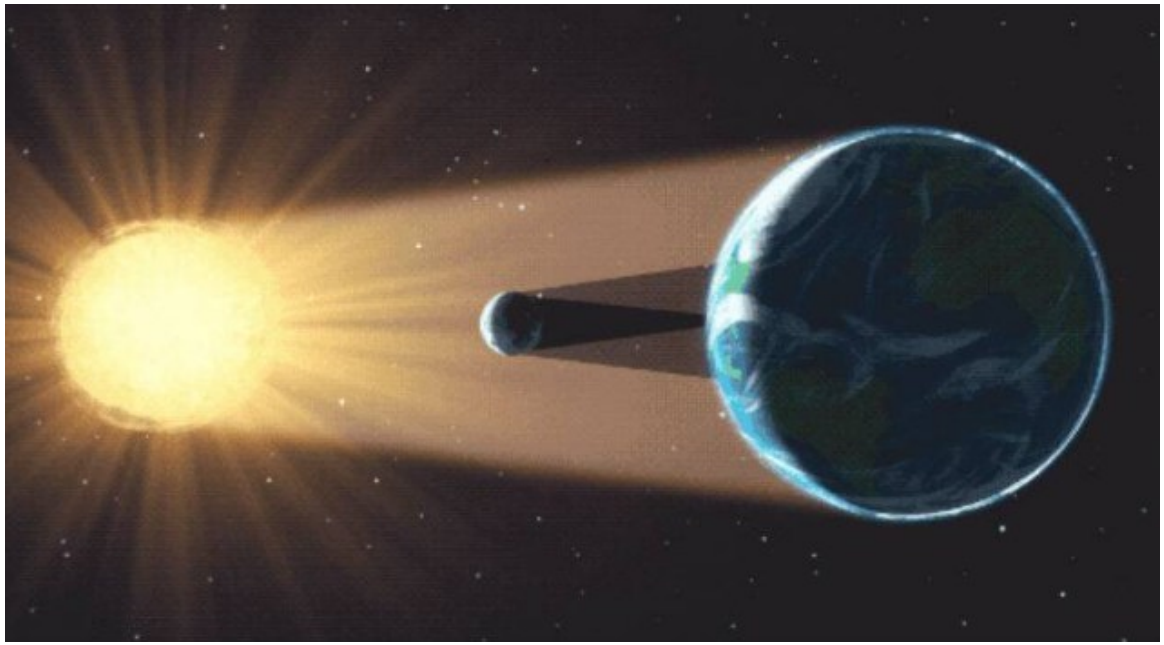
- Planck's law
- Stefan-Boltzmann's law
- Wien's law
- energy conservation



Thermal equilibrium & the black body

A body at temperature T emits radiation as EM waves, that propagates also in vacuum. The higher the temperature, the higher the energy per unit area.

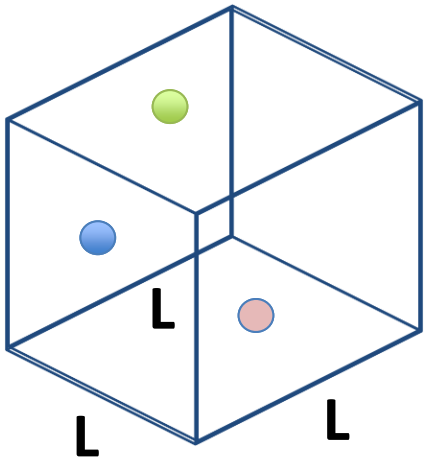
The net radiative balance is the difference between the radiative power absorbed by a body and the radiative power it emits.
At thermal equilibrium, its time-averaged value is zero.

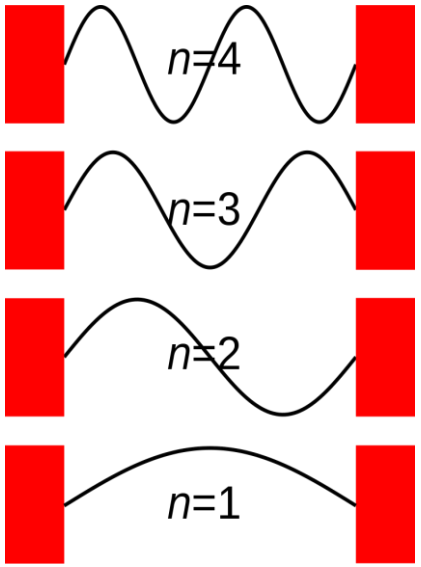


Are Sun, Earth and Moon really a black bodies ?

Emission of power: the particle in a box

Assume the body to be a cubic cavity. Energy can be emitted in EM, with walls imposing EM boundary constraints.





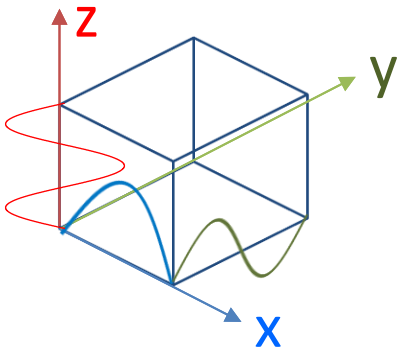
Classic theory

- Each mode is independent from the others
- The energy associated to each is k_B T

Stefan-Boltzmann k_B =1.38 x 10^{-23} J/K

- Each mode corresponds to a frequency:

$$\frac{f}{c} = \frac{1}{\lambda} = \sqrt{f_x^2 + f_y^2 + f_z^2}$$



Increasing L makes the allowed modes more closely spaced and permits longer-wavelength modes.
This causes the number of modes to increase with L^3 and so the power spectrum density

$$D(f) \propto E(f) \cdot df \propto kT_B \cdot f^2$$

Planck-law: power spectrum density

The classical theory predicts a sharp increase of the BB power spectrum with f, not at all consistent with observation (UV catastrophe). Planck assumed that modes are quantized:

energy can be emitted only in **discrete particles**, for a single one (photon):

$$E = hf$$

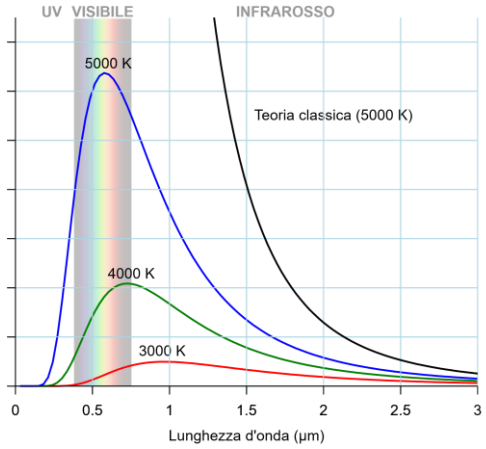
h Planck constant $=6.626 \cdot 10{-}34\ J \cdot s$

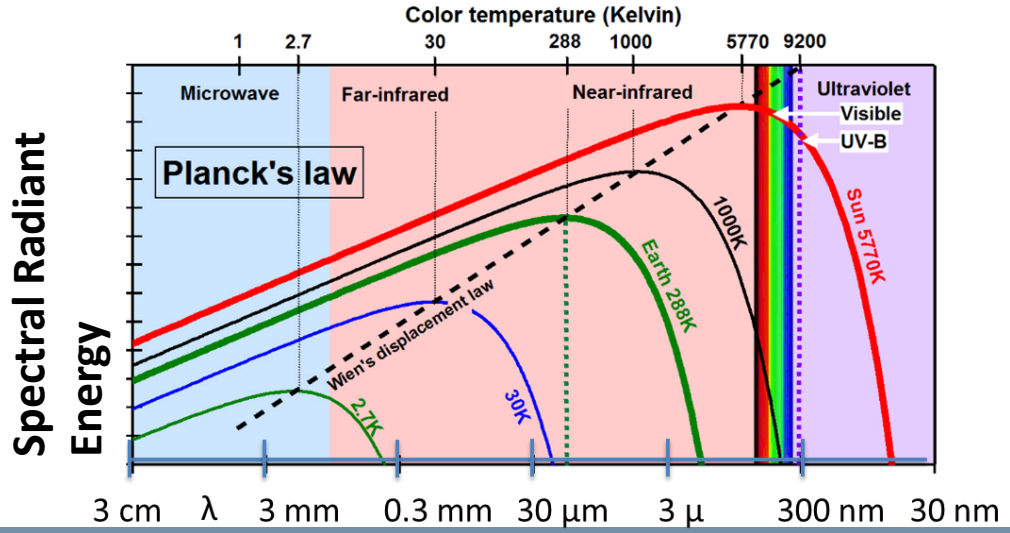
As f increases, the discretization limits the quantity of energy at that frequency (the gaps between photons becomes more and more relevant).

The resultant PSD is:

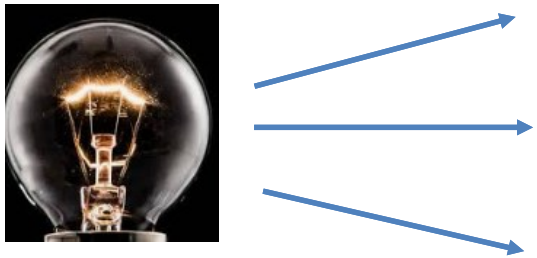
$$L_B(f) = \frac{P(f)}{A_B \cdot df} = \frac{2\pi f^2}{c^2} \frac{hf}{e^{\frac{hf}{kT}} - 1}$$

Spectral Radiant Energy [W/m²/Hz]





Planck power spectrum: Stefan-Boltzmann & Wien

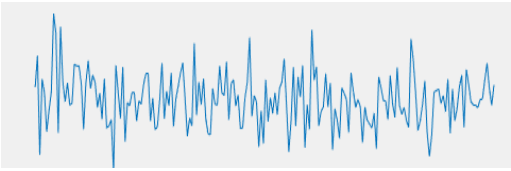


spectral radiant exitance (total radiation leaving the surface)

$$L_B(f) = \frac{2\pi f^2}{c^2} \frac{hf}{e^{\frac{hf}{kT}} - 1} \qquad [W / m^2 / df{=}Hz]$$

$$L_X(\lambda) = \frac{2\pi}{\lambda^5} \frac{hc^2}{e^{\frac{hc}{kT\lambda}} - 1} \qquad [W / m^2 / d\lambda = m]$$

A BB emits the same radiance in any direction.
The radiance does not depend on the observation direction.
The instantaneous signal is modelled as random, zero- mean Gaussian.



The power spectrum depends only on T. We can integrate it to get the total power:

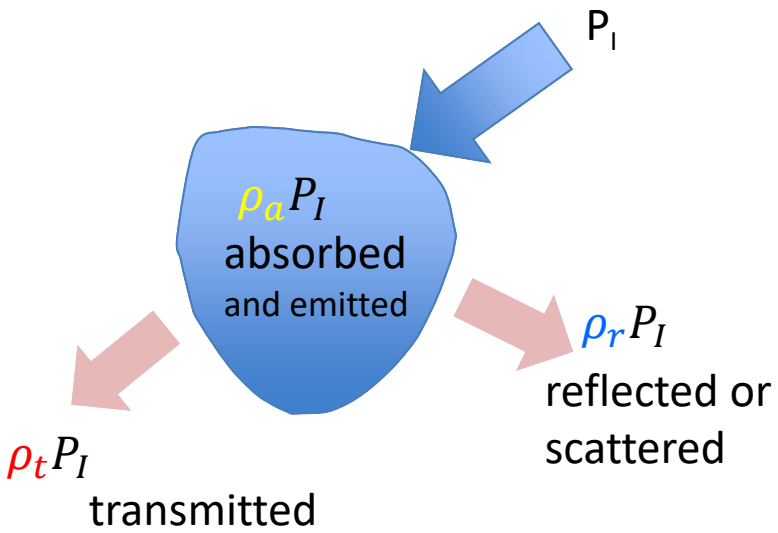
Stefan-Boltzmann law $\frac{P_B}{A_B} = \sigma T^4$

σ=SB constant
=5.67 ·10^{-8} W/m²/K⁴

The peak of the power spectrum is provided by Wien law:

$f_M \simeq 5.87 \cdot 10^{10} \cdot T$ Hz $\qquad \lambda_M \simeq \frac{2898}{T}$ µm

Reflection emission and absorption: energy conservation





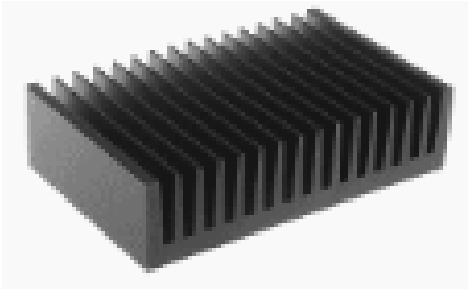




Black, gray and transparent



White dress



Black heatsink



Transparent

Behavior in optics may be different from other ferquencies !

White body: albedo

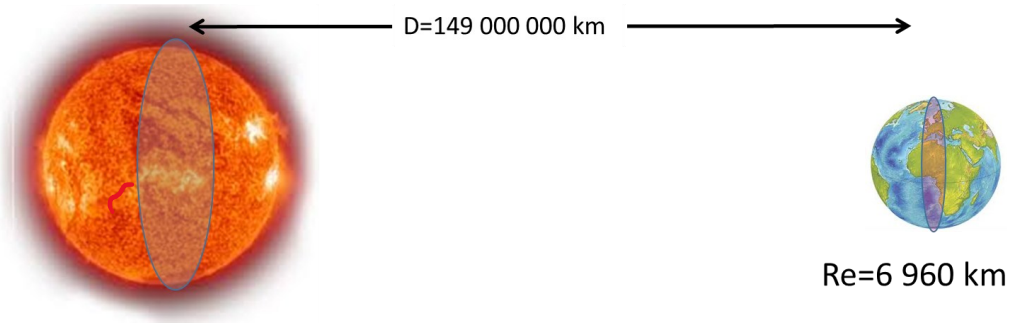
Albedo is the fraction of sunlight that is diffusely reflected by a body. It is measured on a scale from 0 (corresponding to a black body that absorbs all incident radiation) to 1 (corresponding to a body that reflects all incident radiation) [Wikipedia].

Sample albedos

Surface	Typical albedo
Fresh asphalt	0.04[6]
Open ocean	0.06[7]
Worn asphalt	0.12[6]
Conifer forest, summer	0.08,[8] 0.09 to 0.15[9]
Deciduous forest	0.15 to 0.18[9]
Bare soil	0.17[10]
Green grass	0.25[10]
Desert sand	0.40[11]
New concrete	0.55[10]
Ocean ice	0.50 to 0.70[10]
Fresh snow	0.80[10]
Aluminium	0.85[12][13]

Which surfaces absorb the largest fraction of incoming solar radiation and, all else being equal, heat up the most under sunlight?

Example Sun an Earth



Compute the power on a unit cross section at Earth distance.

Compute the Earth's **effective radiative equilibrium temperature**

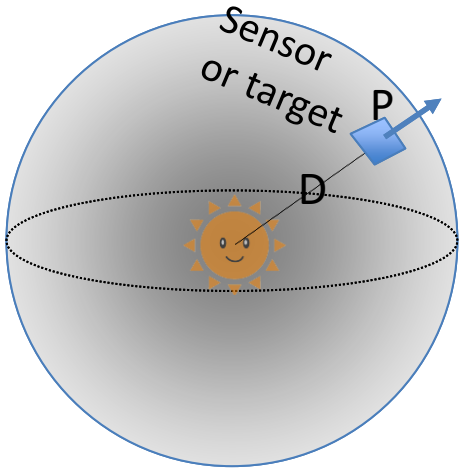
Assume that only 70% of solar radiation is absorbed (the other reflected).

Hint: Earth intercepts radiation over which surface ? but emits over ?

Power received on the cross-section: the irradiance

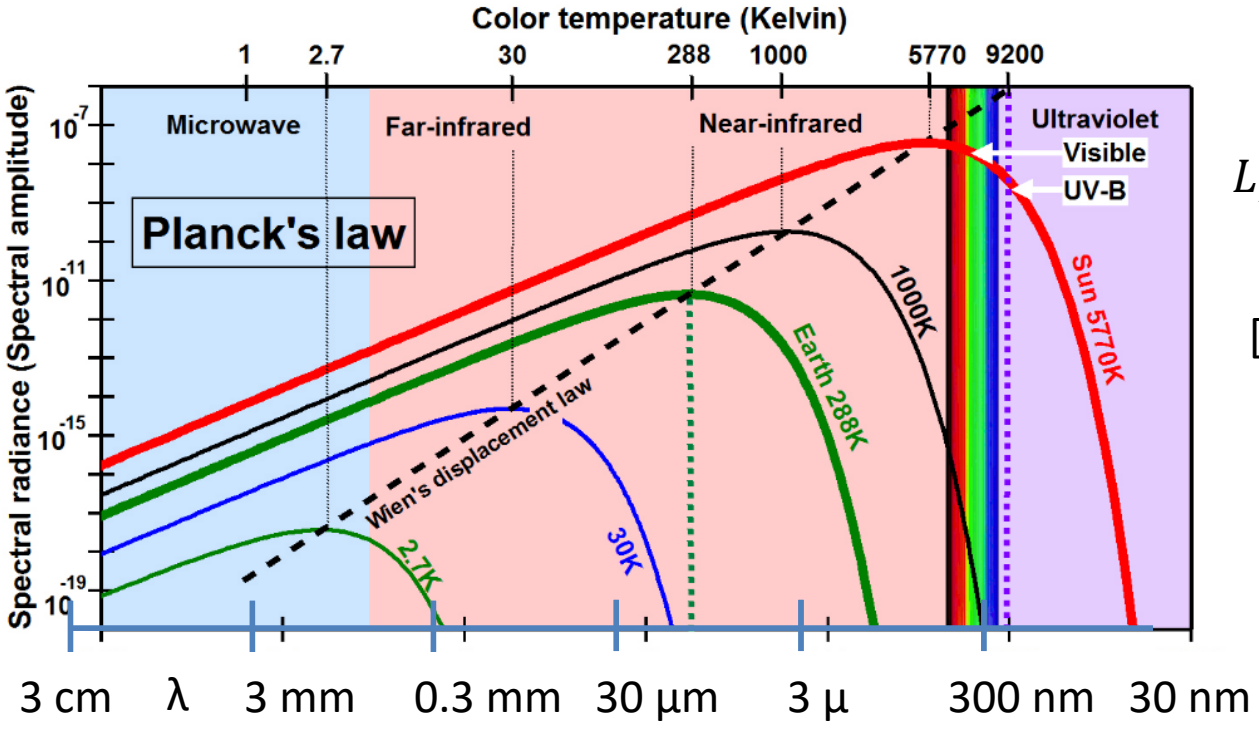
So far, we expressed the [power / BB area].
Yet, we are much more interested to the [power / cross-area at distance D]

The power flux, W/m² on the "receiver" cross-area (orthogonal to the direction of the field):



Planck law: the spectral radiance

Radiance is the radiant power emitted, reflected, transmitted, or received by a surface **per unit projected area and per unit solid angle in a given direction**



$$L_B(f) = \frac{2\pi f^2}{c^2} \frac{hf}{e^{\frac{hf}{kT}} - 1}$$

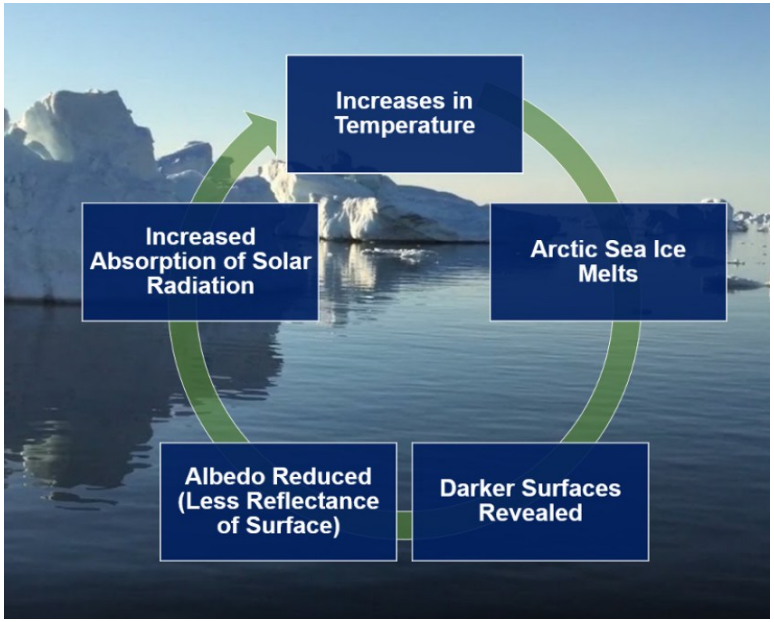
[W/m²/Hz/Sr]

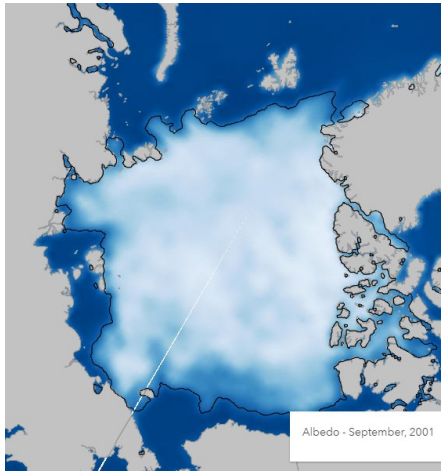
$$L_X(\lambda) = \frac{2}{\lambda^5} \frac{hc^2}{e^{\frac{hc}{kT\lambda}} - 1}$$

[W/m/Sr]

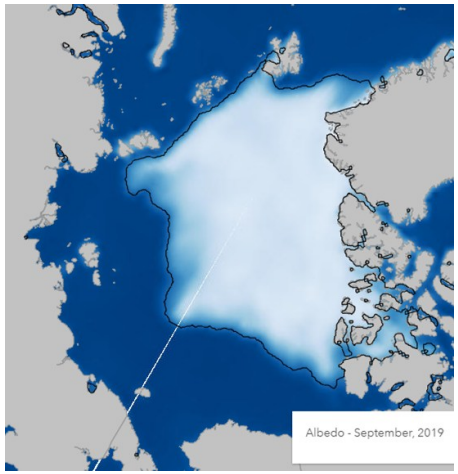
Albedo of ice and water and their impact on climate

Ice reflects much more solar radiation than open water. When ice melts, it exposes darker water, which absorbs more solar energy and promotes further warming and melting: the ice–albedo positive feedback..





Albedo - 2001

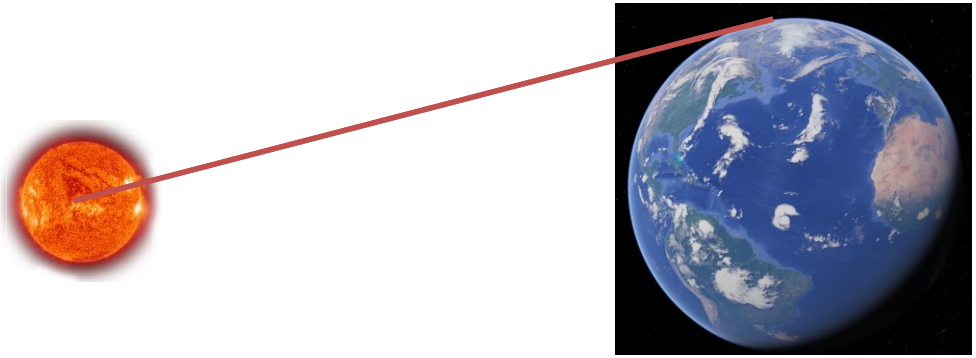


Albedo - 2019

Trend in Sea Ice Percent Coverage

As a region, the Arctic is showing more dramatic signs of climate change than any other part of the planet. These include a warming of air temperatures at a rate two to three times greater than the rest of the planet, and the loss of September sea ice extent at a rate of 13 percent per decade.

Example: compute temperature variation



Example: Cosmic background radiation

The **Cosmic Background**

Explorer (**COBE**) was able to measure in 1989 the radiation spectrum of the cosmic background that fitted Planck low with such an accuracy that error-bars cannot be seen. It is the most precisely measured BB spectrum ever.

The background temperature of:

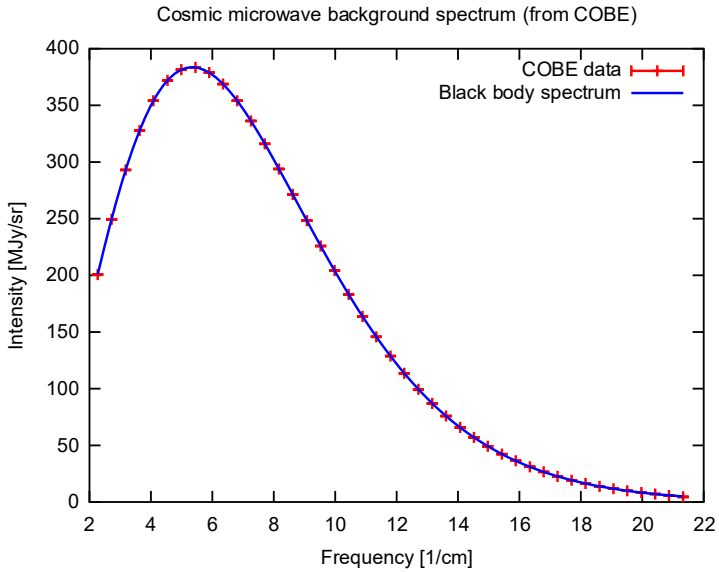
2.72 K (peak at 160 GHz)

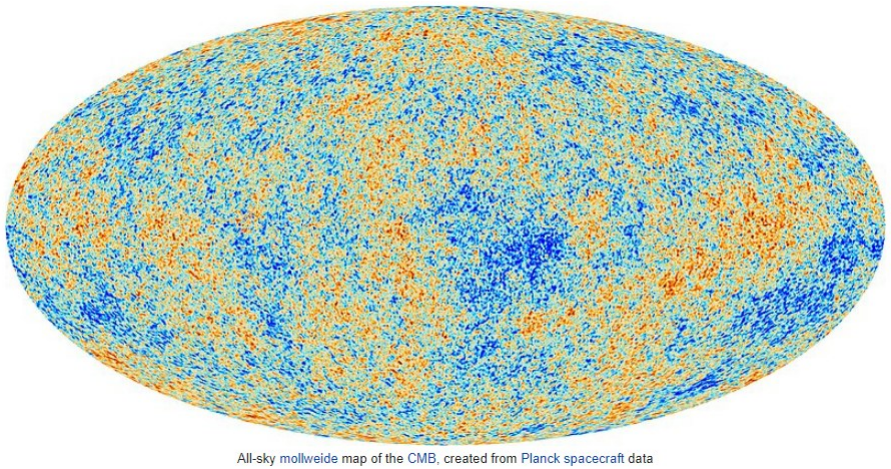
is the evidence of

perfect BB spectrum

(universe was hot and in thermal equilibrium)

that after Big Bang initiated expanding and colding (still on-going).





All-sky mollweide map of the CMB, created from Planck spacecraft data

Cosmic background from Planck space telescope